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Multidisciplinary Design Project

# **Forecasting Earnings Surprises with Artificial Intelligence Techniques**

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## SYMBOL LIST

|            |                                         |
|------------|-----------------------------------------|
| $SUE_i$    | Standardized Unanticipated Earnings     |
| $EPS_i$    | Actual reported Earnings Per Share      |
| $E$        | Mean consensus analyst forecast         |
| $\sigma_i$ | Standard deviation of analyst forecasts |

## **ABBREVIATION LIST**

|       |                                                         |
|-------|---------------------------------------------------------|
| EPS   | Earnings Per Share                                      |
| PEAD  | Post-Earnings Announcement Drift                        |
| EMH   | Efficient Market Hypothesis                             |
| AI    | Artificial Intelligence                                 |
| ARIMA | Autoregressive Integrated Moving Average                |
| BERT  | Bidirectional Encoder Representations from Transformers |
| GRU   | Gated Recurrent Units                                   |
| LSTM  | Long Short-Term Memory                                  |
| MD&A  | Management Discussion & Analysis                        |
| ML    | Machine Learning                                        |
| NLP   | Natural Language Processing                             |
| RF    | Random Forests                                          |
| RNN   | Recurrent Neural Networks                               |
| SEC   | Securities and Exchange Commission                      |
| SUE   | Standardized Unanticipated Earnings                     |
| SVM   | Support Vector Machines                                 |

## **ABSTRACT**

Corporate earnings announcements are critical events that often cause large stock price movements. This project focuses on the Post-Earnings Announcement Drift (PEAD) anomaly, a well-documented market inefficiency where prices continue to drift long after a surprise is announced. The central aim of this study is to develop a quantitative model using Artificial Intelligence (AI) to effectively forecast these earnings surprises.

A key part of our methodology is framing the task as a regression problem: we will predict the precise magnitude (size) of the surprise, not just its simple direction (a "Beat" or "Miss"). This approach is essential for identifying economically significant trading opportunities and filtering out market noise.

The proposed technique is a multimodal deep learning framework. This model will imitate human analysis by fusing two different types of data: (1) structured, time-series data (like historical financial ratios) will be processed by an LSTM network to capture trends, and (2) unstructured, textual data (from 10-Q reports, news articles, and social media) will be processed by FinBERT to analyze sentiment and context.

These two streams will be combined to predict a single, robust target variable: the Standardized Unanticipated Earnings (SUE), which measures the surprise relative to analyst uncertainty.



# 1. INTRODUCTION

This section provides a formal definition of the research topic, establishes its significance within financial markets, defines the project's aim and scope, and reviews the academic literature upon which this study is based. It also defines the core theoretical concepts that serve as the foundation for the proposed methodology.

## 1.1. Background and Motivation

In financial markets, corporate earnings announcements are critical and information-rich events.<sup>1</sup> The disclosure of a firm's quarterly Earnings Per Share (EPS) is a primary indicator for the market to validate or recalibrate its valuation of the firm, frequently leading to significant and immediate price adjustments.

The motivation for this project is to chase two distinct opportunities for generating market-beating returns:

1. **Immediate And High-Volume Price Reaction:** Occurs at the moment the 'Earnings Surprise' (the deviation between actual EPS and consensus analyst expectations) is publicly released.
2. **Post-Earnings Announcement Drift:** The more profound opportunity is an anomaly known as the "*Post-Earnings Announcement Drift*" (PEAD). PEAD is the documented tendency for a stock's price to continue to 'drift' in the direction of the surprise for weeks or even months following the announcement.<sup>2</sup> This persistent drift represents a clear violation of the semi-strong form of the *Efficient Market Hypothesis (EMH)*, which posits that all public information should be reflected in the price near-instantaneously.<sup>2</sup>

The fact that the PEAD anomaly continues to exist suggests the market is not perfectly efficient, which creates a potential opportunity. The main goal of this project is therefore: to build a strong quantitative tool that can separate "market noise"—which means random, unimportant price changes—from the "real, tradable information" hidden in data before the announcement.<sup>4</sup>

## 1.2. Problem Definition, Aim And Scope

This section outlines the specific problem this project addresses, its formal objective, and its operational boundaries.



### 1.2.1. Problem Definition

The central problem in this domain is the accurate forecasting of corporate earnings surprises. Many existing models focus on simple directional classification ("Beat" or "Miss")<sup>5</sup>, an approach that has been shown to be insufficient for developing viable, cost-aware trading strategies.<sup>7</sup>

### 1.2.2. Aim of the Study

The formal aim of this study is: *To design, implement, and evaluate a quantitative model utilizing Artificial Intelligence (AI) techniques to forecast the magnitude of 'Earnings Surprises' for equities in the US market.*<sup>7</sup>

### 1.2.3. Scope of the Study

The scope of this project is defined as follows:

- **Market:** The components of the S&P 500 index. This universe is selected due to its high liquidity, market stability, and, most importantly, the extensive availability of high-quality analyst consensus data (e.g., from I/B/E/S), which is essential for calculating the target variable.<sup>7</sup>
- **Period:** The study will utilize data from 2010 to 2025. This period is deliberately chosen to train and test the model on modern market structures, which are dominated by algorithmic and high-frequency trading. It consciously excludes the unique systemic volatility of the 2008 financial crisis.<sup>7</sup>
- **Methodological Focus:** As outlined in the aim, the project's focus is not merely on classifying the *direction* of a surprise, but on forecasting its precise *numerical magnitude* to identify economically significant events.<sup>4</sup>

## 1.3. Review of Related Works

This section presents a review of the academic literature relevant to the project. The review is structured chronologically to demonstrate the evolution of predictive methodologies, from foundational theories to state-of-the-art techniques.

### 1.3.1. Foundational Studies on Market Efficiency and Anomalies

This research is based on the Efficient Market Hypothesis (EMH). EMH states that asset prices include all information, making it impossible to get extra returns.<sup>10</sup> However, a lot of research challenges this idea.

The most important study for this project is by Ball and Brown (1968).<sup>13</sup> This key paper was the first to show two things: first, that accounting income numbers are useful to the market

<sup>18</sup>; and second, that the market reacts to this information slowly. Ball and Brown saw a continued price "drift" for firms with positive surprises and a negative drift for firms with negative surprises.<sup>19</sup>

This discovery of the PEAD anomaly <sup>20</sup> showed that market reactions are slow. This provides the main academic reason for this project.<sup>2</sup>

### 1.3.2. Econometric and Classical Machine Learning Approaches

Early models used methods like ARIMA (Autoregressive Integrated Moving Average) to predict earnings.<sup>21</sup> The main problem was that these models assume financial data is simple and linear, which it is not.

The first AI models in finance used classical machine learning (ML), like Support Vector Machines (SVM) and Random Forests (RF), on *structured financial data*.<sup>21</sup> Their main benefit was finding complex, *non-linear patterns* that simple regression models could not.<sup>5</sup> Studies showed these ML models were better than older models and sometimes even better than human analysts.<sup>5</sup>

### 1.3.3. State-of-the-Art: Deep Learning and Alternative Data

Current top methods involve two advances: using deep learning for time-series data and using Natural Language Processing (NLP) for unstructured "alternative" data.

Deep learning models like Recurrent Neural Networks (RNNs), specifically LSTM and GRU, were the next big step.<sup>25</sup> LSTMs are better than static models (like RF) because they are built to find patterns in *sequences* of data. An LSTM can look at a *sequence* of financial ratios over time, finding patterns like seasonality that static models miss.<sup>28</sup> Studies show LSTMs are more accurate for EPS prediction than analysts and other models.<sup>28</sup>

At the same time, a major change happened: models started analyzing *text*, not just *numbers*. This means using NLP to find predictive signals in unstructured data like 1.0-K/1.0-Q reports, news, and earnings call transcripts.<sup>1</sup> This text includes forward-looking information that is not in the financial numbers.

This project will use a special NLP model for finance. General models (like BERT trained on Wikipedia) do not work well because financial language is unique.<sup>33</sup> Models like **FinBERT** are pre-trained on many financial documents (like 1.0-Ks). They understand financial context better and are much better at financial sentiment tasks than general models.<sup>34</sup> While some studies in smaller markets (e.g., Poland) found simpler models were just as good <sup>38</sup>, the proof for using models like FinBERT in the US market is strong.

## 2. PROPOSED METHODOLOGY

This section outlines the proposed technique for the design project. The methodology is a direct synthesis of the findings from the literature review, designed to represent a state-of-the-art approach to financial forecasting.

### 2.1. Key Concepts And Theoretical Framework

This section defines the core key concepts for the project, with a specific focus on the precise definition of the target variable.

The project's success depends on a good definition of "Earnings Surprise." A simple "raw surprise" (e.g.,  $EPS_{Actual} - EPS_{Consensus}$ ) is uninformative, as a \$0.01\$ surprise is meaningless without context. A "percentage surprise" ( $Surprise - EPS_{Consensus}$ ) is statistically problematic and highly unstable, as it is prone to extreme errors when the denominator (consensus EPS) is near zero.<sup>8</sup>

Therefore, this project will use a stronger, academically preferred metric: **Standardized Unanticipated Earnings (SUE)**. The concept was operationalized in foundational research on the PEAD anomaly.<sup>39</sup>

The SUE for a firm  $i$  is formally defined as:

$$SUE_i = \frac{EPS_i - E}{\sigma_i}$$

The components of this equation are:

- **Numerator ( $EPS_i - E$ ):** The "raw surprise," calculated as the actual reported EPS  $EPS_i$  minus the mean consensus analyst forecast ( $E$ ).<sup>40</sup>
- **Denominator ( $\sigma_i$ ):** The *standard deviation* of all individual analyst forecasts for firm  $i$  during the period.<sup>40</sup>

The denominator,  $\sigma_i$ , is the critical component. It is a measure of analyst uncertainty (how much the analysts disagree).<sup>8</sup> The superiority of the SUE metric is best illustrated by two examples:

1. **Case 1 (Low Uncertainty):** Twenty analysts all forecast an EPS of \$1.00\$ ( $\sigma_i$  is near zero). The firm reports \$1.10\$. The SUE value will be *massive*, as the firm delivered a result that was far outside the confident consensus. This is a *true*, statistically significant surprise.

2. **Case 2 (High Uncertainty):** Ten analysts forecast \$0.50\$ and ten forecast \$1.50\$. The mean consensus  $E$  is still \$1.00\$. The firm again reports \$1.10\$. In this case, the denominator ( $\sigma_i$ ) is huge. The resulting SUE value will be *tiny*. This is a *false* surprise; it is just noise because the analysts were not sure.

By dividing the raw surprise by the level of analyst uncertainty, the SUE metric automatically finds surprises that are *statistically significant*. It highlights events that beat a strong agreement and ignores events that fall inside a wide, uncertain range. This makes SUE the perfect target variable for a model that needs to find "tradable information".<sup>4</sup>

## 2.2. Task Formulation: Regression Vs. Classification

We must make an important distinction about this predictive task. Many previous studies tries to forecast *direction* of the surprise (e.g., a "Beat" or "Miss").<sup>95</sup> But only the direction is not an enough forecast for a trustable trade strategy. A classification model treats a small, unimportant surprise (e.g., actual EPS of \$1.21\$ vs. consensus of \$1.20\$) the same as a massive, very significant surprise (e.g., actual EPS of \$1.50\$ vs. consensus of \$1.20\$). Both are just called a "Beat".<sup>7</sup>

Therefore, this project will frame the task as a *regression* problem: predicting the magnitude of the surprise. This approach is essential for the project's goal. It allows the model to filter out small-sized (and low-confidence) predictions that are likely to be "eaten" by transaction costs and market noise.<sup>4</sup> A successful regression model allows a strategy to focus only on large-sized, highly significant forecasts that have real economic potential.<sup>48</sup>

## 2.3. A Multimodal Predictive Framework

**Table 2.2** Evolution of earnings surprise prediction models

| Approach and Era                                   | Data Type                      | Example                                 | Key Limitation / Next Step                            |
|----------------------------------------------------|--------------------------------|-----------------------------------------|-------------------------------------------------------|
| <b>Foundational</b> (1960s-1980s)                  | Simple Time-Series             | Linear Regression <sup>14</sup>         | Too simple; misses complex patterns.                  |
| <b>Econometric</b> (1980s-2000s)                   | Time-Series EPS                | ARIMA, GARCH <sup>21</sup>              | Still linear; ignores other firm data.                |
| <b>Classical ML</b> (2015s-Present)                | Structured (Ratios)            | Random Forest, SVM <sup>5</sup>         | Good at non-linear patterns, but static; misses time. |
| <b>Deep Learning (Temporal)</b><br>(2015s-Present) | Structured (Time-Series)       | LSTM, GRU <sup>28</sup>                 | Sees time, but blind to text/sentiment.               |
| <b>Deep Learning (Textual)</b><br>(2018s-Present)  | Unstructured (Text)            | FinBERT <sup>30</sup>                   | Sees text, but blind to time-series numbers.          |
| <b>Proposed Multimodal</b>                         | <b>Hybrid (Numbers + Text)</b> | <b>Fused LSTM-FinBERT <sup>45</sup></b> | <b>Combines numbers and textual data</b>              |

As shown in the literature review and **Table 2.2**, modern research has two main paths: (1) using deep learning (like LSTMs) for *structured* number data, and (2) using NLP (like FinBERT) for *unstructured* text data. A model that uses only one of these paths is missing a lot of predictive information.

The proposed method is therefore a *multimodal deep learning framework*. This is a good approach because the two data types (numbers and text) *complete* each other; they are not repetitive. A human analyst reads *both* a balance sheet (numbers) and an earnings call (text) to get the full story.<sup>46</sup> For instance, the number data (Modality 1) might show falling sales, but the text data (Modality 2) might explain *why*, pointing to a new product that will reverse this trend.

This combined approach (fusing numbers and text) is noted in recent academic papers as a top method for financial forecasting.<sup>45</sup>

## 2.4. Data Modalities And Feature Engineering

The model will be trained on two distinct data modalities, which will be engineered into features for the neural network.

### 2.4.1. Modality 1: Structured (Tabular) Time-Series Data

This stream will consist of quantitative time-series data for each firm.

- **Data Source:** Publicly available financial datasets (e.g., Compustat, CRSP) or financial data provider APIs.
- **Features:** A time-series vector will be constructed for each firm, incorporating data from the preceding 8-16 quarters. This vector will include:
  1. **Fundamental Ratios:** Key financial health indicators (e.g., Price-to-Earnings, Return on Assets, Debt-to-Equity, gross/net profit margins).
  2. **Market Data:** Historical stock volatility, average trading volume, and momentum indicators.
  3. **Consensus Data:** Analyst consensus forecast ( $E$ ), standard deviation of forecasts ( $\sigma_i$ ), and historical SUE values (as an auto-regressive feature).<sup>40</sup>

### 2.4.2. Modality 2: Unstructured (Textual) Data

This stream will consist of qualitative data extracted from a wide range of public sources.

- **Data Source:** This includes official documents like 10-K (annual) and 10-Q (quarterly) reports filed with the SEC, and transcripts from quarterly earnings conference calls.<sup>30</sup> To broaden the analysis, this stream will also include alternative text sources such as

financial news articles, company press releases, and sentiment data from social media posts.<sup>20</sup>

- **Feature Engineering (NLP):** The text from the "Management Discussion & Analysis" (MD&A) section of 10-Qs and the "Forward-Looking Statements" portion of earnings calls will be processed using a pre-trained **FinBERT** model.<sup>34</sup>
- **Features:** The FinBERT model will be used to generate two sets of features from the text:
  1. **Sentiment Scores:** A numerical value (positive, negative, neutral) representing the tone of the text.<sup>30</sup>  
**Semantic Embeddings:** A high-dimensional vector (e.g., a [768x1] vector) that represents the full *semantic meaning* and context of the management discussion. This embedding is far more information-rich than a simple sentiment score.<sup>34</sup>

## 2.5. Proposed Model Architecture

The proposed method is a multimodal neural network. It has two parallel 'streams' that process data separately, an idea based on recent research.<sup>45</sup>

1. **Stream 1 (For Numbers):** An **LSTM (or GRU) network** will be used to process the structured time-series data (Modality 1).<sup>28</sup> Because this is a 'recurrent' network, it is designed to learn patterns over time, like trends and seasonality, from the company's past financial numbers.
2. **Stream 2 (For Text):** A pre-trained **FinBERT** model will be used to process the unstructured text data (Modality 2).<sup>34</sup> This stream will read the text and output a vector (embedding) that captures the text's meaning and the management's sentiment.
3. **Fusion Layer:** The output vectors from Stream 1 (numbers) and Stream 2 (text) will be combined (or 'fused') into one single, wide vector.<sup>45</sup> This combined vector represents the model's total understanding of both the firm's numbers and its text-based context.

**Output Layer (Regression):** This final combined vector will be fed into a few standard 'Dense' layers. These layers will narrow down to a **single output neuron**. This final neuron is trained (using Mean Squared Error) to predict one single number: the **Standardized Unanticipated Earnings (SUE)**.<sup>40</sup>

### 3. IMPLEMENTATION AND EXPERIMENTAL SETUP

Following the theoretical framework established in the previous section, this chapter details the practical implementation of the proposed multimodal architecture. The focus during this phase has been on establishing the development environment, constructing the data ingestion pipelines for both modalities, and implementing the core neural network modules.

#### 3.1. Development Environment and Tools

The model is being developed using Python 3.10, leveraging the PyTorch framework for deep learning tasks due to its dynamic computation graph and extensive support for NLP transformers.

- **Data Acquisition:** yfinance library is utilized to fetch historical market data (Open, High, Low, Close, Volume) and pandas for time-series manipulation. For textual data, the SEC EDGAR database and publicly available financial corpus datasets are accessed.
- **NLP Backend:** The transformers library by Hugging Face is used to implement the pre-trained **FinBERT** model.
- **Hardware:** Preliminary training and testing are conducted on a cloud-based GPU environment (T4/P100) to accelerate the fine-tuning process of the BERT layers.

#### 3.2. Data Preprocessing Pipeline

A robust preprocessing pipeline is essential for multimodal learning. We have successfully implemented two parallel processing streams:

##### 3.2.1. Processing Modality 1: Structured Time-Series

Raw financial data often contains noise and missing values. The following steps have been applied:

1. **Normalization:** To ensure the LSTM converges efficiently, all financial ratios and price data are normalized using MinMaxScaler to scale values between (0, 1).
2. **Sequence Generation:** A sliding window approach has been coded to create input sequences of length  $T = 16$  (representing 16 quarters of historical data) to predict the earnings surprise of the  $(T + 1)^{th}$  quarter.

##### 3.2.2. Processing Modality 2: Unstructured Text

For the textual stream, raw text from quarterly reports requires tokenization compatible with FinBERT.

1. **Tokenization:** We utilize the BertTokenizer pre-trained on the 'ProsusAI/finbert' corpus.

2. **Truncation and Padding:** Sequences are standardized to a maximum length of 512 tokens. Texts exceeding this limit are truncated to preserve the most critical sections (typically the introduction and conclusion of MD&A sections), while shorter texts are padded with [PAD] tokens.

## 4. PRELIMINARY ARCHITECTURE IMPLEMENTATION

In this phase, the theoretical model defined in Section 2.5 has been translated into executable code. The architecture is modularized into three distinct Python classes.

### 4.1. The LSTM Encoder (Numeric Stream)

A custom TimeDistributed LSTM module has been implemented. This module takes the tensor of shape (Batch\_Size, Sequence\_Length, Num\_Features) and outputs a hidden state vector representing the temporal financial context.

- *Implementation Status:* The LSTM class is fully defined with configurable hidden dimensions and dropout layers to prevent overfitting.

### 4.2. The FinBERT Encoder (Text Stream)

We have integrated the BertModel from the transformers library. The weights are frozen during the initial epochs to retain the financial linguistic knowledge learned during pre-training, allowing only the top classification layers to update.

- *Implementation Status:* The text encoding pipeline successfully converts raw text into 768-dimensional embeddings.

### 4.3. Multimodal Fusion Network

The core innovation of this project, the Fusion Layer, has been constructed. This layer concatenates the output vector from the LSTM (Numeric) and the [CLS] token embedding from FinBERT (Text).

$$V_{fusion} = [ V_{LSTM} \oplus V_{FinBERT} ]$$

This combined vector is then passed through a Fully Connected (Dense) network to regress the final SUE value. Initial dry-run tests confirm that the data shapes align correctly across both streams, allowing for end-to-end gradient flow.



## 5. CURRENT PROGRESS AND NEXT STEPS

### 5.1. Achievements

- **Methodology Mastery:** Successfully integrated NLP and Time-Series libraries (PyTorch, Transformers).
- **Data Pipeline:** Automated fetching and cleaning of S&P 500 OHLCV data.
- **Model Architecture:** The complete MultimodalEarningsPredictor class is coded and compiles without errors.

### 5.2. Future Work

The final phase will focus on the full-scale training of the model. This includes hyperparameter tuning (learning rate, batch size) and comparative analysis against baseline models (ARIMA and Random Forest). The final output will be the experimental results demonstrating the Mean Squared Error (MSE) improvement over unimodal baselines.

## REFERENCES

1. Extracting the Structure of Press Releases for Predicting Earnings Announcement Returns, erişim tarihi Kasım 8, 2025, <https://arxiv.org/html/2509.24254v1>
2. (PDF) A review of the Post-Earnings-Announcement Drift, erişim tarihi Kasım 8, 2025, [https://www.researchgate.net/publication/347976957\\_A\\_review\\_of\\_the\\_Post-Earnings-Announcement\\_Drift](https://www.researchgate.net/publication/347976957_A_review_of_the_Post-Earnings-Announcement_Drift)
3. A review of the Post-Earnings-Announcement Drift - IDEAS/RePEc, erişim tarihi Kasım 8, 2025, <https://ideas.repec.org/a/eee/beexfi/v29y2021ics2214635020303750.html>
4. Assessing the Effects of Earnings Surprise on ... - Duke Economics, erişim tarihi Kasım 8, 2025, [https://public.econ.duke.edu/~get/browse/courses/201/spr12/2010-PRESENTATIONS/FullECON202-FINAL\\_PAPERS/2009-December/Lim\\_Thesis.pdf](https://public.econ.duke.edu/~get/browse/courses/201/spr12/2010-PRESENTATIONS/FullECON202-FINAL_PAPERS/2009-December/Lim_Thesis.pdf)
5. Predicting Future Earnings Changes Using Machine ... - NYU Stern, erişim tarihi Kasım 8, 2025, <https://www.stern.nyu.edu/sites/default/files/assets/documents/SSRN-id3741015.pdf>
6. A Large Scale Multi-modal Benchmark for Earning Surprise Prediction - arXiv, erişim tarihi Kasım 8, 2025, <https://arxiv.org/html/2510.03965v1>
7. Streaks in Earnings Surprises and the Cross-Section of Stock Returns - Chapman University Digital Commons, erişim tarihi Kasım 8, 2025, [https://digitalcommons.chapman.edu/cgi/viewcontent.cgi?article=1116&context=business\\_articles](https://digitalcommons.chapman.edu/cgi/viewcontent.cgi?article=1116&context=business_articles)
8. DD-050-Matematik Mühendisliği bitirme Çalışması ve Matematik Mühendisliğinde Tasarım Uygulaması Hazırlama Esasları.pdf
9. How to measure earnings surprises: Based on revised market ... - NIH, erişim tarihi Kasım 8, 2025, <https://pmc.ncbi.nlm.nih.gov/articles/PMC10745228/>
10. THE EFFICIENT MARKET HYPOTHESIS ON TRIAL, erişim tarihi Kasım 8, 2025, <https://www.westga.edu/~bquest/2002/market.htm>
11. The Efficient Market Hypothesis and Its Critics by Burton G. Malkiel, Princeton University CEPS Working Paper No. 91, erişim tarihi Kasım 8, 2025, <https://www.princeton.edu/~ceps/workingpapers/91malkiel.pdf>
12. Redalyc.THE EFFICIENT MARKET HYPOTHESIS: A CRITICAL REVIEW OF LITERATURE AND METHODOLOGY, erişim tarihi Kasım 8, 2025, <https://www.redalyc.org/pdf/6922/692273691006.pdf>
13. Ball & Brown (1968) | PDF | Cyberspace | Communication - Scribd, erişim tarihi Kasım 8, 2025, <https://www.scribd.com/document/644308442/Ball-Brown-1968>
14. Empirical Evaluation Of Accounting Income Numbers - IDEAS/RePEc, erişim tarihi Kasım 8, 2025, <https://ideas.repec.org/a/bla/joares/v6y1968i2p159-178.html>
15. Ball, R. and Brown, P. (1968) Empirical Evaluation of Accounting Income Numbers. Journal of Accounting Research, 6, 159-178. - References - Scientific Research Publishing, erişim tarihi Kasım 8, 2025, <https://www.scirp.org/reference/referencespapers?referenceid=2404068>
16. Why a 1968 Paper Still Influences Modern Asset Management - Knowledge at Wharton, erişim tarihi Kasım 8, 2025,

- <https://knowledge.wharton.upenn.edu/podcast/knowledge-at-wharton-podcast/jacobs-levy-award-2019/>
17. [PDF] An empirical evaluation of accounting income numbers - Semantic Scholar, erişim tarihi Kasım 8, 2025, <https://www.semanticscholar.org/paper/An-empirical-evaluation-of-accounting-income-Ball-Brown/4d9595491bb8e21c527828e8483bc612bceaf773>
  18. A REEXAMINATION OF BALL AND BROWN - Journal of Management and Innovation, erişim tarihi Kasım 8, 2025, <https://jmi.mercy.edu/index.php/JMI/article/view/119/69>
  19. The Strengths and Weaknesses of the 1968 'Ball and Brown' Study | UKEssays.com, erişim tarihi Kasım 8, 2025, <https://www.ukessays.com/essays/finance/review-of-the-ball-and-brown-study-finance-essay.php>
  20. Ball and Brown (1968) After Five Decades, erişim tarihi Kasım 8, 2025, <https://jacobslevycenter.wharton.upenn.edu/wp-content/uploads/2019/09/Ball-and-Brown-Presentation.pdf>
  21. A Review of ARIMA vs. Machine Learning Approaches for Time ..., erişim tarihi Kasım 8, 2025, <https://www.mdpi.com/1999-5903/15/8/255>
  22. A bibliometric literature review of stock price forecasting: From statistical model to deep learning approach - NIH, erişim tarihi Kasım 8, 2025, <https://pmc.ncbi.nlm.nih.gov/articles/PMC10943735/>
  23. Comparison of SVM and ARIMA Model in Stock Market - Atlantis Press, erişim tarihi Kasım 8, 2025, <https://www.atlantispress.com/article/125983649.pdf>
  24. Man versus Machine Learning: The Term Structure of Earnings Expectations and Conditional Biases | The Review of Financial Studies | Oxford Academic, erişim tarihi Kasım 8, 2025, <https://academic.oup.com/rfs/article/36/6/2361/6782974>
  25. Forecasting S&P 500 Using LSTM Models - arXiv, erişim tarihi Kasım 8, 2025, <https://arxiv.org/html/2501.17366v1>
  26. [2201.08218] Long Short-Term Memory Neural Network for Financial Time Series - arXiv, erişim tarihi Kasım 8, 2025, <https://arxiv.org/abs/2201.08218>
  27. Advanced Stock Market Prediction Using Long Short-Term Memory Networks: A Comprehensive Deep Learning Framework - arXiv, erişim tarihi Kasım 8, 2025, <https://arxiv.org/html/2505.05325v1>
  28. Earnings Prediction with Deep Learning, erişim tarihi Kasım 8, 2025, <https://arxiv.org/abs/2006.03132>
  29. Advanced Deep Learning Techniques for Analyzing Earnings Call Transcripts: Methodologies and Applications - arXiv, erişim tarihi Kasım 8, 2025, <https://arxiv.org/html/2503.01886v1>
  30. LEVERAGING TEXT MINING TO EXTRACT INSIGHTS FROM EARNINGS CALL TRANSCRIPTS - AllianceBernstein, erişim tarihi Kasım 8, 2025, <https://www.alliancebernstein.com/content/dam/global/insights/insights-whitepapers/leveragingtextminingtoextractinsights-chinfan.pdf>
  31. Using AI to unlock investment and risk management opportunities in earnings call transcripts, erişim tarihi Kasım 8, 2025, <https://www.lseg.com/en/insights/data-analytics/ai-unlock-investment-risk-management-opportunities-earnings-call-transcripts>

32. Assessing the Predictive Power of Earnings Call Transcripts on Next-Day Stock Price Movement: A Semantic Analysis Using Large Language Models | UC Berkeley School of Information, erişim tarihi Kasım 8, 2025, <https://www.ischool.berkeley.edu/projects/2024/assessing-predictive-power-earnings-call-transcripts-next-day-stock-price-movement>
33. Financial Sentiment Analysis Using FinBERT with Application in Predicting Stock Movement, erişim tarihi Kasım 8, 2025, <https://arxiv.org/html/2306.02136v3>
34. (PDF) FinBERT: A Large Language Model for Extracting Information from Financial Text†, erişim tarihi Kasım 8, 2025, [https://www.researchgate.net/publication/364070191\\_FinBERT\\_A\\_Large\\_Language\\_Model\\_for\\_Extracting\\_Information\\_from\\_Financial\\_Text](https://www.researchgate.net/publication/364070191_FinBERT_A_Large_Language_Model_for_Extracting_Information_from_Financial_Text)
35. Fine-Tuned FinBERT Model with Sentiment Focus Method for Enhancing Sentiment Analysis of FOMC Minutes - Alexandria (UniSG), erişim tarihi Kasım 8, 2025, <https://www.alexandria.unisg.ch/server/api/core/bitstreams/1d94cc0d-30b9-4d0d-9131-8e8c20c46837/content>
36. FinBERT: A Pre-trained Financial Language Representation Model for Financial Text Mining - IJCAI, erişim tarihi Kasım 8, 2025, <https://www.ijcai.org/proceedings/2020/0622.pdf>
37. Comparative Investigation of GPT and FinBERT's Sentiment Analysis Performance in News Across Different Sectors - MDPI, erişim tarihi Kasım 8, 2025, <https://www.mdpi.com/2079-9292/14/6/1090>
38. Are Natural Language Processing methods applicable to EPS ..., erişim tarihi Kasım 8, 2025, <https://www.aimspress.com/article/doi/10.3934/DSFE.2025003?viewType=HTML>
39. Standardized Unexpected Earnings - QuantConnect.com, erişim tarihi Kasım 8, 2025, <https://www.quantconnect.com/research/15369/standardized-unexpected-earnings/>
40. The Impact of Individual and Collective Attribution on Earnings Calls ..., erişim tarihi Kasım 8, 2025, <https://repository.upenn.edu/bitstreams/2926ada1-fbba-4abc-9f55-1ccc9a08ccc6/download>
41. (PDF) Borrower Distress and the Efficiency of Relationship Banking - ResearchGate, erişim tarihi Kasım 8, 2025, [https://www.researchgate.net/publication/322345322\\_Borrower\\_Distress\\_and\\_the\\_Efficiency\\_of\\_Relationship\\_Banking](https://www.researchgate.net/publication/322345322_Borrower_Distress_and_the_Efficiency_of_Relationship_Banking)
42. A New Measure of Earnings surprises and Post-Earnings-Announcement Drift - Brandeis, erişim tarihi Kasım 8, 2025, <https://peeps.unet.brandeis.edu/~heidifox/ese.pdf>
43. Full article: Beyond the street EPS surprise – when 'other surprises' matter in explaining earnings announcement returns, erişim tarihi Kasım 8, 2025, <https://www.tandfonline.com/doi/full/10.1080/00014788.2024.2400875>
44. Pre-Earnings Announcement Over-Extrapolation - Mendoza College of Business, erişim tarihi Kasım 8, 2025, [https://mendoza.nd.edu/wp-content/uploads/2019/01/2016\\_spring\\_finance\\_seminar\\_series\\_peter\\_kelly\\_paper\\_updated\\_3\\_22\\_2016.pdf](https://mendoza.nd.edu/wp-content/uploads/2019/01/2016_spring_finance_seminar_series_peter_kelly_paper_updated_3_22_2016.pdf)
45. Financial Forecasting from Textual and Tabular Time Series - ACL ..., erişim tarihi

- Kasım 8, 2025, <https://aclanthology.org/2024.findings-emnlp.486/>
46. A Large Scale Multi-modal Benchmark for Earning Surprise ... - arXiv, erişim tarihi Kasım 8, 2025, <https://arxiv.org/pdf/2510.03965>
47. Financial Forecasting from Textual and Tabular Time Series - ACL Anthology, erişim tarihi Kasım 8, 2025, <https://aclanthology.org/2024.findings-emnlp.486.pdf>
48. Standardized Unexpected Earnings - University of West Georgia, erişim tarihi Kasım 8, 2025, <https://www.westga.edu/~bquest/2002/unexpected.htm>
49. Dong Liang. Predicting Stock Price Changes with Earnings Call Transcripts. A Master's - Carolina Digital Repository, erişim tarihi Kasım 8, 2025, <https://cdr.lib.unc.edu/downloads/j67317338>
50. Stock Price Prediction Using FinBERT-Enhanced Sentiment with SHAP Explainability and Differential Privacy - MDPI, erişim tarihi Kasım 8, 2025, <https://www.mdpi.com/2227-7390/13/17/2747>

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